A LARGER EXPLICIT ZERO-FREE REGION FOR THE RIEMANN ZETA-FUNCTION

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Abstract. We prove that the Riemann zeta-function has no zeros in the region $\sigma \geq 1 - 1/(4.81406 \log |t|)$, $|t| \geq 2$. This improves the constant 4.896 of Bellotti, Trudgian and Yang. We follow their method with several changes. The test function is chosen separately for each range of heights. The positivity bound for pairs of zeros is sharper. The remainder in Kadiri's explicit formula is written as a sum over the trivial zeros, which gives a bound that does not grow with the height. The smoothing kernel is a polynomial on its support, so its Laplace transform is known exactly. The finite inequalities of the proof are checked by computer in exact rational and interval arithmetic.

1. Introduction

De la Vallée Poussin proved in 1899 that the Riemann zeta-function $\zeta(s)$ has no zeros in a region

$$\sigma \geq 1 - \frac{1}{R \log |t|}, \qquad |t| \geq 2, \qquad s = \sigma + it, \tag{1.1}$$

for some constant $R > 0$. Such a region gives the error term $\pi(x) - \mathrm{Li}(x) = O(x \exp(-\sqrt{\log x / R}))$ in the prime number theorem [5]. A smaller R gives a better explicit error term for large x, so one asks how small R can be taken.

Kadiri proved (1.1) with $R = 5.69693$ [5], and Mossinghoff, Trudgian and Yang lowered this to $R = 5.558691$ [7]. The best published value is $R = 4.896$ for $t \geq 3$, due to Bellotti, Trudgian and Yang [1]. We refer to their paper as BTY. They also obtain $R = 4.8594$ if one uses a constant from Yang's thesis, 19.62 in place of 21.233 in (2.2) below.

Theorem 1.1. *Let $R = 240703/50000 = 4.81406$. Then $\zeta(\sigma + it) \neq 0$ whenever $|t| \geq 2$ and $\sigma \geq 1 - 1/(R \log |t|)$.*

The new constant is smaller than both 4.896 and 4.8594. We use no zero-free region stronger than Yang's (2.2), and we use the Riemann hypothesis only up to height $3 \cdot 10^{12}$. A finite list of inequalities in the proof is checked by computer. Section 9 gives this list and the arithmetic used.

Earlier work. Our argument is that of BTY, which builds on several earlier ideas. Stechkin used the zeros of ζ in pairs ρ, $1 - \bar{\rho}$, symmetric about the critical line [10]. Heath-Brown smoothed the sum $\sum \Lambda(n) n^{-s}$ with a compactly supported kernel whose Laplace transform has nonnegative real part [2]. Kadiri combined the two ideas [5], and her smoothed explicit formula is our starting point. BTY introduced the test function $\eta w(\eta u) \kappa(e^{-(2\sigma-1)u})$. Here $1 - \eta$ is the real part of a hypothetical zero, w is a smoothing kernel, and κ is a polynomial close to $1/(1+x)$ on $[0, 1]$. We follow BTY in the rest of the argument as well. Pairs of zeros give a nonnegative contribution, one pair of zeros is isolated, and distant zeros are handled with

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Lehman's lemmas [6]. The gamma factor and the pole are bounded separately. BTY also bound the prime-power sum from below by grouping the powers of each small prime. Finally, we use the verification of the Riemann hypothesis by Platt and Trudgian [9] and Yang's zero-free region of Littlewood type [13], which limits the heights we have to treat.

What is new. We change BTY's argument in six places.

(1) *Height bands.* We first fix the height of a hypothetical minimal zero. Then we choose the multiplier κ and the other constants for the band of heights that contains it (Section 3).
(2) *Paired positivity.* BTY bound the positive and the negative terms of a rational function with different denominators. We keep the common scale $2\sigma - 1$, so all terms have the same denominator and nothing is lost to the signs (Proposition 5.3).
(3) *Remainder.* The remainder in Kadiri's formula equals an absolutely convergent sum over the trivial zeros $-2, -4, \dots$ of ζ (Proposition 6.3). The earlier bound for it grew like the logarithm of the height. The new bound is of order η^3 and does not depend on the height. Each shifted explicit formula also gets its own lower bound for the real part of its argument (Lemma 6.5).
(4) *Kernel.* The kernel is the autocorrelation of a quartic polynomial. It is a polynomial of degree 9 on its support, so its Laplace transform has a finite exact formula (Lemma 4.4).
(5) *Prime powers.* Each prime power keeps its own weight, which comes from the kernel, the multiplier and an upper bound σ^+ for σ. Interval arithmetic gives lower bounds for the minima of the resulting cosine sums (Lemma 7.5).
(6) *Final check.* The final inequality is checked in interval arithmetic on the whole closed parameter domain, boundary included (Proposition 8.3).

The trigonometric polynomial (degree 12) and the multipliers (degrees 10 and 14) are also new.

Plan of the paper. Section 2 fixes notation and lists the results we use. Section 3 reduces Theorem 1.1 to ruling out a minimal zero in one of seven height bands. Sections 4 and 5 construct the kernel and the test function and prove paired positivity. Section 6 proves the identity for Kadiri's remainder, and Section 7 bounds the other terms of the explicit formula. Section 8 combines the bounds and proves Theorem 1.1. Section 9 describes the computer check, and Appendix A lists all parameters.

2. Notation and inputs from the literature

2.1. **Notation.** In (1.1), $\sigma + it$ is a generic point. From Section 3 on, σ and t are fixed numbers attached to a hypothetical zero, and generic complex variables are s and z. All logarithms are natural. The letter $\rho = \beta + i\gamma$ runs over the nontrivial zeros of ζ, counted with multiplicity. We write $\psi = \Gamma'/\Gamma$ for the digamma function and Λ for the von Mangoldt function. The integral $\int_{(1/2)}$ is taken upwards along the line $\operatorname{Re} z = 1/2$. We fix

$$R = \frac{240703}{50000}, \qquad A_* = \frac{1}{R} = \frac{50000}{240703}, \qquad H = 3 \cdot 10^{12}, \qquad T_0 = 10^{10}, \qquad K = 12.$$

Here H is the height of the Riemann hypothesis verification, T_0 is an offset in the cutoff for distant zeros, and K is the degree of the trigonometric polynomial. A decimal number with finitely many digits denotes an exact rational. Numbers ending in "..." and numbers called rounded are approximations, given only for orientation.

2.2. **Inputs from the literature.** We quote each input in our notation. Apart from the standard facts listed at the end of this subsection, we use no other result about ζ.

(K1) Kadiri [5, Théorème 1.1]. $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1 - 1/(R_0 \log|\operatorname{Im} s|)$ and $|\operatorname{Im} s| \geq 2$, where $R_0 = 5.69693$.

(K2) Kadiri [5, Proposition 2.1, eq. (7)]. Let $d' > 0$. Let $g \geq 0$ be supported in $[0, d')$, of class C^2 on $[0, d']$, with $g(d') = g'(0) = g'(d') = g''(d') = 0$. For a function φ on $[0,\infty)$ write $\mathcal{L}\varphi(z) = \int_0^\infty e^{-zu}\varphi(u)\,\mathrm{d}u$ for its Laplace transform. Then, for every $s \in \mathbb{C}$ with $\operatorname{Re} s > 1/2$, we have

(2.1)
$$\operatorname{Re}\sum_{n\geq 1}\frac{\Lambda(n)}{n^s}g(\log n) = g(0)\Big(-\frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{s}{2}+1\Big)\Big) + \operatorname{Re}\mathcal{L}g(s-1) - \sum_\rho \operatorname{Re}\mathcal{L}g(s-\rho) + \operatorname{Re}\Big\{\frac{1}{2\pi i}\int_{(1/2)}\frac{\mathcal{L}g''(s-z)}{(s-z)^2}\operatorname{Re}\psi\Big(\frac{z}{2}\Big)\,\mathrm{d}z + \frac{\mathcal{L}g''(s)}{s^2}\Big\}.$$

Kadiri states (2.1) for all $s \in \mathbb{C}$, but we use it only for $\operatorname{Re} s > 1/2$. She first proves it for $\operatorname{Re} s > 1$ (pp. 314–317) and then extends it by one argument on p. 317. Both sides are harmonic functions of s defined beyond $\operatorname{Re} s > 1$, so the identity persists. For $\operatorname{Re} s > 1/2$ this is the identity theorem for harmonic functions on a connected open set. The left side is a finite sum, because g has compact support, so it is harmonic in s everywhere. On the right, each term is harmonic for $\operatorname{Re} s > 1/2$, because the sum over ρ converges locally uniformly and the integrand has no singularity on the line $\operatorname{Re} z = 1/2$. The remainder in the last line has a plus sign.

(K3) Kadiri [5, eq. (28)]. For $x > 0$ and $y^2 > x^2$,
$$\operatorname{Re}\psi(x+iy) = \log|y| - \frac{x}{2(x^2+y^2)} + r(x,y), \qquad |r(x,y)| \leq \frac{1}{12x|y|} + \frac{x^2}{2y^2}.$$

(PT) Platt and Trudgian [9, Theorem 1]. Every zero ρ with $0 < \gamma \leq 3\cdot 10^{12}$ has $\beta = 1/2$.

(Y) Yang [13, Corollary 1.2]. For $|t| \geq 3$,
$$\zeta(\sigma+it) \neq 0 \qquad \text{whenever} \qquad \sigma > 1 - \frac{\log\log|t|}{21.233\log|t|}. \tag{2.2}$$

(L1) Lehman [6, Lemma 1]. Let ϕ be continuous, positive and decreasing on $[T_1, T_2]$, where $2\pi e \leq T_1 < T_2$. Then, with γ running over the positive ordinates of zeros,
$$\sum_{T_1<\gamma\leq T_2}\phi(\gamma) = \frac{1}{2\pi}\int_{T_1}^{T_2}\phi(x)\log\frac{x}{2\pi}\,\mathrm{d}x + \vartheta\Big\{4\phi(T_1)\log T_1 + 2\int_{T_1}^{T_2}\frac{\phi(x)}{x}\,\mathrm{d}x\Big\}, \qquad |\vartheta| \leq 1.$$

(L2) Lehman [6, Lemma 2]. If $T \geq 2\pi e$, then $\sum_{\gamma>T}\gamma^{-2} < T^{-1}\log T$.

(D) The digamma function [8, 5.7.6, 5.5.2]. For $z \notin \{0, -1, -2, \dots\}$,
$$\psi(z) = -\gamma_E + \sum_{k=0}^{\infty}\Big(\frac{1}{k+1} - \frac{1}{k+z}\Big), \qquad \psi(z+1) = \psi(z) + \frac{1}{z},$$
where γ_E is Euler's constant. In particular ψ is meromorphic with simple poles of residue -1 at $0, -1, -2, \dots$, and ψ increases on $(0,\infty)$.

We also use standard facts. The nontrivial zeros satisfy $0 < \beta < 1$, and ζ has no zeros on the real segment $(0,1)$. Every bounded region contains only finitely many zeros of ζ, because $(s-1)\zeta(s)$ is entire and not identically zero. By the functional equation, the zeros are symmetric

under $\rho \mapsto 1-\bar\rho$ and $\rho \mapsto \bar\rho$, with multiplicities. We use the minimum principle for harmonic functions, the residue theorem and Sturm's theorem.

Bernstein coefficients. On an interval $[\alpha_1, \alpha_2]$, the Bernstein basis of degree n consists of the polynomials

$$\chi_{n,l}(u) = \binom{n}{l}\Big(\frac{u-\alpha_1}{\alpha_2-\alpha_1}\Big)^l\Big(\frac{\alpha_2-u}{\alpha_2-\alpha_1}\Big)^{n-l}, \qquad 0 \le l \le n.$$

They are nonnegative on the interval and sum to 1. Every polynomial g of degree at most n has a unique expansion $g = \sum_l g_l^{\mathrm{B}}\chi_{n,l}$. The numbers g_l^{B} are its Bernstein coefficients of degree n. Hence $\min_l g_l^{\mathrm{B}} \le g \le \max_l g_l^{\mathrm{B}}$ on the interval. Raising the degree from n to $n+1$ replaces the coefficients by convex combinations of them, so these bounds can only get sharper.

3. Reduction to a minimal zero in a height band

This section turns Theorem 1.1 into a statement about one hypothetical zero with parameters in an explicit compact set.

Lemma 3.1 (Minimal zero)**.** *Suppose that Theorem 1.1 is false. Then the minimum A of $(1-\beta)\log\gamma$ over the zeros $\rho = \beta + i\gamma$ with $\beta > 1/2$ and $\gamma > 0$ exists. Every zero $\rho_0 = \beta_0 + it$ attaining it satisfies*

$$t > H, \qquad \frac{1}{6} < A \le A_*, \qquad L_0 < \log t < L_7,$$

where $L_0 = 28.7296$ and $L_7 = 82.32067$.

Proof. Take a zero $\rho_1 = \beta_1 + i\gamma_1$ with $|\gamma_1| \ge 2$ and $\beta_1 \ge 1 - 1/(R\log|\gamma_1|)$. Replacing ρ_1 by $\bar\rho_1$ if needed, we may assume $\gamma_1 > 0$. Since $R\log 2 > 2$, we have $\beta_1 > 1/2$, and $(1-\beta_1)\log\gamma_1 \le A_*$. We call a zero with $\beta > 1/2$ and $\gamma > 0$ admissible. In particular, ρ_1 is admissible.

By (PT), zeros with $0 < \gamma \le H$ have $\beta = 1/2$, so they are not admissible. Hence every admissible zero has $\gamma > H$. By (Y), every such ρ satisfies $(1-\beta)\log\gamma \ge \log\log\gamma/21.233$. Hence the admissible zeros with $(1-\beta)\log\gamma \le A_*$ lie in a bounded region. There are finitely many of them, and ρ_1 is one of them, so the minimum exists and $A \le A_*$.

By (K1), every admissible zero has $(1-\beta)\log\gamma > 1/5.69693 > 1/6$. For a minimizer, $\log\log t \le 21.233A \le 21.233A_* < \log L_7$, where the last inequality is check (C1) of Section 9. Finally $\log t > \log H > L_0$, again by (C1). □

From now on $\rho_0 = \beta_0 + it$ is a minimizing zero. We use the following quantities:

$$\begin{gathered} L = \log t, \qquad \eta = 1-\beta_0 = \frac{A}{L}, \qquad T = Kt + T_0, \qquad \sigma = 1 - \frac{A}{\log T}, \\ h = 2\sigma - 1, \qquad \mu = \frac{1-\sigma}{\eta} = \frac{L}{\log(Ke^L + T_0)}, \qquad s_k = \sigma + ikt \quad (0 \le k \le K). \end{gathered} \tag{3.1}$$

The ratio μ depends on L only. We call η the depth of ρ_0 and h the strip width. The height T is the cutoff between near and distant zeros in Lemma 7.1. Every test point s_k has height at most Kt, so a zero above T is at vertical distance more than T_0 from all of them.

3.1. The trigonometric polynomial.

The trigonometric polynomial is

$$P(x) = \frac{\big|\sum_{l=0}^{12} c_l e^{ilx}\big|^2}{\sum_{l=0}^{12} c_l^2} = \sum_{k=0}^{12} a_k\cos(kx), \qquad a_+ = \sum_{k=1}^{12} a_k, \tag{3.2}$$

with the integers c_l of (A.1) in the appendix. Check (C3) computes all a_k exactly and shows that $a_0 = 1$ and $a_k \geq 0$. Clearly $P \geq 0$. The two values used below are

$$a_1 = \frac{579010044340076601089576}{333333333333509945567445} = 1.73703\ldots,$$
$$a_+ = \frac{697654317962413463864378}{200000000000105967340467} = 3.48827\ldots.$$

The same P is used in all bands.

3.2. **Height bands.** We split $[L_0, L_7]$ into seven closed bands $[L_{j-1}, L_j]$, $1 \leq j \leq 7$, with

$$(L_0, \ldots, L_7) = (28.7296,\ 45,\ 60,\ 70,\ 77,\ 80,\ 82,\ 82.32067).$$

The band rectangle is $U_j = [L_{j-1}, L_j] \times [1/6, A_*]$. By Lemma 3.1, (L, A) lies in at least one band rectangle, and we fix one such j.

From here on we also treat (L, A) as a free point of U_j. The quantities (3.1) are then defined by the same formulas with $t = e^L$. Lemma 3.2 and the certificate concern all of U_j, and the minimal zero gives one point of it. All later constants with a superscript or an index belong to this band. Each band has rational constants σ^-, η^+ and ε, listed in Table 1. The bands and all parameters were found numerically. The proof uses only the fact that they pass the checks of Section 9. We also put

$$\sigma^+ = 1 - \frac{1/6}{\log(Ke^{L_j} + T_0)}, \qquad h^- = 2\sigma^- - 1, \qquad h^+ = 2\sigma^+ - 1, \qquad \sigma_m^- = \sigma^- + h^- m. \tag{3.3}$$

Lemma 3.2 (Band bounds)**.** *Let $(L, A) \in U_j$, and define σ, η, h and μ by (3.1) with $t = e^L$. Then*

$$\tfrac{1}{2} < \sigma^- < \sigma \leq \sigma^+ < 1, \qquad 0 < \eta < \eta^+ < 1, \qquad h^- < h \leq h^+ < 1, \qquad \tfrac{9}{10} < \mu < 1.$$

Proof. We have $\log T = \log(Ke^L + T_0)$, which increases with L, and $\sigma = 1 - A/\log T$. With $A \leq A_*$ and $L \geq L_{j-1}$ we get $\sigma \geq 1 - A_*/\log(Ke^{L_{j-1}} + T_0)$, and this exceeds σ^- by check (C5). With $A \geq 1/6$ and $L \leq L_j$ we get $\sigma \leq \sigma^+$. Next $\eta = A/L \leq A_*/L_{j-1} < \eta^+$ by (C5), and (C5) also gives $1/2 < \sigma^- < \sigma^+ < 1$ and $\eta^+ < 1$. The bounds for h follow. The bound $\mu < 1$ holds because $\log(Ke^L + T_0) > L$. For the lower bound, write $\mu = L/(L + \log(K + T_0 e^{-L}))$. The logarithm decreases in L, so μ increases in L. At $L = L_0$ we have $T_0 e^{-L_0} < 1$ and $\log 13 < L_0/9$, so $\mu > 9/10$. The program also checks $9/10 < \mu < 1$ on every box (check (C13)). □

Lemma 3.3 (Zeros below the cutoff)**.** *Every zero $\rho = \beta + i\gamma$ with $|\gamma| \leq T$ satisfies $1 - \sigma \leq \beta \leq \sigma$.*

Proof. If $|\gamma| \leq H$, then $\beta = 1/2$ by (PT), symmetry and the absence of real zeros in $(0, 1)$. Also $1 - \sigma < 1/2 < \sigma$. Let $H < |\gamma| \leq T$ and $\beta \geq 1/2$. If $\beta = 1/2$ there is nothing to prove. Otherwise ρ or $\bar{\rho}$ is admissible, so $(1 - \beta)\log|\gamma| \geq A$ by minimality. Hence $\beta \leq 1 - A/\log|\gamma| \leq 1 - A/\log T = \sigma$. If $\beta < 1/2$, apply this to the zero $1 - \bar{\rho}$ to get $1 - \beta \leq \sigma$. □

4. The kernel

This section defines the smoothing kernel w and proves the properties of its Laplace transform that the argument needs. The kernel is the same in all bands.

Definition 4.1 (Kernel)**.** Let

$$b = -\frac{2409073}{20000000} = -0.12045365, \qquad d = \frac{52698243}{50000000} = 1.05396486.$$

Let $q(y) = (1-y^2)(1+by^2)$ for $|y| \le 1$ and $q(y) = 0$ otherwise. Put $\|q\|_2^2 = \int_{-1}^{1} q(y)^2 \,\mathrm{d}y$ and

$$w(u) = \frac{10}{\|q\|_2^2} \int_{\mathbb{R}} q(x)\, q\Big(x - \frac{2u}{d}\Big) \,\mathrm{d}x \qquad (u \in \mathbb{R}), \qquad W(z) = \int_0^\infty e^{-zu} w(u) \,\mathrm{d}u.$$

We write $w_0 = w(0)$ and $W_0(z) = W(z) - w_0/z$.

Thus $w(u)$ is a multiple of the autocorrelation of q at lag $2u/d$. Equivalently, it is a multiple of the autocorrelation of $x \mapsto q(2x/d)$ at lag u. Its support in $[0,\infty)$ is $[0,d]$. The value $\|q\|_2^2 = 8116714872719329/7875000000000000$ is computed exactly. Let w_4^* be the largest absolute value of the degree-13 Bernstein coefficients of w'''' on $[0,d]$. It is an exact rational, $w_4^* = 797.74\ldots$. The polynomial w'''' has degree 5. We raise its degree to 13 because this gives a sharper bound (Section 2.2). We use the two constants

$$C(\nu) = |w''(0)| + \frac{w'''(0)(1+e^{-\nu d})}{\nu} + \frac{w_4^*}{\nu^2} \quad (\nu > 0), \qquad C_- = |w''(0)| + d\, e^d (w'''(0) + w_4^* d). \tag{4.1}$$

The function $C(\nu)$ decreases in ν.

Lemma 4.2 (Kernel)**.** (1) *The function w is even and vanishes for $|u| \ge d$. On $[0,d]$ it equals a polynomial of degree 9 with rational coefficients, and $w_0 = 10$.*

(2) *$0 \le w(u) \le w_0$ for all u, and w is nonincreasing on $[0,\infty)$.*

(3) *$w'(0) = 0$, $w(d) = w'(d) = w''(d) = 0$, $w'''(d) = -w'''(0)$, and $w''(0) < 0 < w'''(0)$. The derivatives at d are those of the polynomial piece. In particular w is of class C^2 on $[0,\infty)$.*

(4) *$\operatorname{Re} W(iy) \ge 0$ for every $y \in \mathbb{R}$.*

(5) *$|w''''(u)| \le w_4^*$ for $0 \le u \le d$.*

Proof. (1) The integrand vanishes unless $|x| < 1$ and $|x - 2u/d| < 1$, which forces $|u| < d$. Evenness follows from the substitution $x \mapsto x + 2u/d$. For $0 \le v \le 2$ the integral $\int_{v-1}^{1} q(x)q(x-v)\,\mathrm{d}x$ is a polynomial in v of degree at most 9. Its rational coefficients are computed exactly in check (C2), and the degree is 9. At $u = 0$ the integral equals $\|q\|_2^2$, so $w_0 = 10$.

(2) Nonnegativity holds because $q \ge 0$, and the bound $w \le w_0$ is the Cauchy–Schwarz inequality. For monotonicity, note that $q = 1 + (b-1)y^2 - by^4$ on $[-1,1]$. As a function of $y^2 \in [0,1]$, its derivative is linear, with values $b-1$ and $-1-b$ at the endpoints, both negative. So q is even, continuous, takes values in $[0,1]$, and decreases strictly in $|y|$ on $[-1,1]$. Hence each superlevel set $\{q > \theta\}$ with $0 \le \theta < 1$ is an open interval $(-\alpha(\theta), \alpha(\theta))$ centred at 0. By the layer-cake formula, $q(x) = \int_0^1 \mathbf{1}[q(x) > \theta]\,\mathrm{d}\theta$. Insert this for both factors in the integral defining w and use Tonelli's theorem (all integrands are nonnegative):

$$\int_{\mathbb{R}} q(x)q(x-v)\,\mathrm{d}x = \int_0^1 \int_0^1 \ell_{\theta,\theta'}(v)\,\mathrm{d}\theta\,\mathrm{d}\theta', \qquad v = \frac{2u}{d},$$

where $\ell_{\theta,\theta'}(v)$ is the length of $(-\alpha(\theta),\alpha(\theta)) \cap (v - \alpha(\theta'), v + \alpha(\theta'))$. Two intervals with half-lengths α_1, α_2 and centres at distance $|v|$ overlap in length $\max\{0, \min(2\alpha_1, 2\alpha_2, \alpha_1 + \alpha_2 - |v|)\}$. If one interval contains the other, the overlap is $2\min(\alpha_1,\alpha_2)$. Otherwise it is $\alpha_1 + \alpha_2 - |v|$ or 0. This length is nonincreasing in $|v|$, so each $\ell_{\theta,\theta'}(v)$ is nonincreasing in $v \ge 0$. Hence $w(u)$ is nonincreasing in $u \ge 0$.

(3) These identities and signs are checked in exact rational arithmetic (check (C2)). The extension by zero is C^2 because w, w' and w'' vanish at d.

(4) Let $\hat{q}(x) = \int_{\mathbb{R}} q(y) e^{-ixy}\,\mathrm{d}y$ be the Fourier transform of q. Since w is even and real,

$$\operatorname{Re} W(iy) = \int_0^\infty \cos(yu)\, w(u)\,\mathrm{d}u = \frac{1}{2}\int_{\mathbb{R}} e^{-iyu} w(u)\,\mathrm{d}u.$$

Substitute $u = dv/2$ and then $v = x - x'$ in the double integral defining w. With $\xi = dy/2$ this gives

$$\int_{\mathbb{R}} e^{-iyu} w(u)\,\mathrm{d}u = \frac{10}{\|q\|_2^2} \cdot \frac{d}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-i\xi x} q(x)\, e^{i\xi x'} q(x')\,\mathrm{d}x\,\mathrm{d}x' = \frac{5d}{\|q\|_2^2}\,|\hat{q}(\xi)|^2,$$

since q is real. Hence $\operatorname{Re} W(iy) = (5d/(2\|q\|_2^2))\,|\hat{q}(dy/2)|^2 \geq 0$.

(5) The Bernstein basis polynomials of degree 13 on $[0, d]$ are nonnegative and sum to 1. Hence $|w''''|$ is at most the largest absolute Bernstein coefficient, which is w_4^*. □

Lemma 4.3 (Transform bounds)**.** *The function W is entire. If $\operatorname{Re} z \geq \nu > 0$, then $|W_0(z)| \leq C(\nu)|z|^{-3}$. If $\operatorname{Re} z \geq -1$ and $z \neq 0$, then $|W_0(z)| \leq C_-|z|^{-3}$.*

Proof. W is entire because w is bounded with compact support. Integrate by parts three times and use $w(d) = w'(0) = w'(d) = w''(d) = 0$:

$$W(z) = \frac{w_0}{z} + \frac{1}{z^3}\Big(w''(0) + \int_0^d e^{-zu} w'''(u)\,\mathrm{d}u\Big).$$

One more integration by parts gives

$$\int_0^d e^{-zu} w'''(u)\,\mathrm{d}u = \frac{w'''(0) - e^{-zd} w'''(d)}{z} + \frac{1}{z}\int_0^d e^{-zu} w''''(u)\,\mathrm{d}u.$$

For $\operatorname{Re} z \geq \nu > 0$ we have $|z| \geq \nu$, $|w'''(d)| = w'''(0)$ and $\int_0^d e^{-\nu u}\,\mathrm{d}u \leq 1/\nu$. This gives the first bound. For $\operatorname{Re} z \geq -1$ we use $|e^{-zu}| \leq e^d$ and $|w'''(u)| \leq w'''(0) + w_4^* u$ on $[0, d]$. This gives $|\int_0^d e^{-zu} w'''(u)\,\mathrm{d}u| \leq d\,e^d(w'''(0) + w_4^* d)$ and the second bound. □

Lemma 4.4 (Exact transform)**.** *For every $z \neq 0$,*

$$W(z) = \sum_{l=0}^{9} \frac{w^{(l)}(0) - e^{-zd}\, w^{(l)}(d)}{z^{l+1}}, \tag{4.2}$$

where $w^{(l)}(d)$ are the derivatives of the polynomial piece at d.

Proof. On $[0, d]$, w is a polynomial of degree 9, so $w^{(10)} = 0$ there. Ten integrations by parts on $[0, d]$ give (4.2) with no remainder. □

The derivatives $w^{(l)}(0)$ and $w^{(l)}(d)$ are exact rationals. The program evaluates (4.2) at real arguments with interval arithmetic.

5. The test function and paired positivity

This section defines the test function of BTY with our kernel and band-dependent multiplier. Its main result is a positivity bound for pairs of zeros in the strip $0 \leq \operatorname{Re} z \leq h$.

Definition 5.1 (Multiplier and test function)**.** For the fixed band, let $\kappa(x) = \sum_{m=0}^{M} \kappa_m x^m$ be the polynomial with the rational coefficients of Table 3 or 4. Here $\kappa_0 = 1$, $M = 10$ in bands 1–6 and $M = 14$ in band 7. Put $\|\kappa\|_1 = \sum_m |\kappa_m|$ and $\tilde{\kappa}_m = \kappa_m + \kappa_{m-1}$ for $0 \leq m \leq M + 1$, with $\kappa_{-1} = \kappa_{M+1} = 0$. Define

$$w_\eta(u) = \eta\, w(\eta u), \qquad f(u) = w_\eta(u)\,\kappa(e^{-hu}),$$

$$F(z) = \int_0^\infty e^{-zu} f(u)\,\mathrm{d}u = \sum_{m=0}^{M} \kappa_m W\Big(\frac{z + hm}{\eta}\Big),$$

and the paired transform $\Pi(z) = \operatorname{Re} F(z) + \operatorname{Re} F(h - \bar{z})$.

The numbers $\tilde\kappa_m$ are the coefficients of $(1+x)\kappa(x)$. The multiplier κ is close to $1/(1+x)$ on $[0,1]$, which is the reason for its form [1].

Lemma 5.2 (Multiplier)**.** *The multiplier κ is positive and strictly decreasing on $[0,1]$. Consequently $f \ge 0$.*

Proof. Write $\kappa = \sum_{l=0}^{M} \kappa_l^{\mathrm{B}} \chi_{M,l}$ in the Bernstein basis of degree M on $[0,1]$. Check (C4) shows that all κ_l^{B} are positive and $\kappa_{l+1}^{\mathrm{B}} < \kappa_l^{\mathrm{B}}$ for every l. Positive coefficients give $\kappa > 0$ on $[0,1]$. The derivative κ' has Bernstein coefficients $M(\kappa_{l+1}^{\mathrm{B}} - \kappa_l^{\mathrm{B}}) < 0$, so $\kappa' < 0$. Finally $f \ge 0$ because $w \ge 0$ and $e^{-hu} \in (0,1]$. □

The function Π satisfies $\Pi(h-\bar z) = \Pi(z)$. Since w is real, $F(\bar z) = \overline{F(z)}$, so $\operatorname{Re} F(h-\bar z) = \operatorname{Re} F(h-z)$ and Π is harmonic.

Proposition 5.3 (Paired positivity)**.** *Assume check (C6) for the band, and let $h \in [h^-, 1)$ and $\eta \in (0, \eta^+]$. Then $\Pi(z) \ge 0$ for all z with $0 \le \operatorname{Re} z \le h$.*

Proof. The line $\operatorname{Re} z = 0$. For $z = iy$ we have $h - \bar z = h + iy$, so

$$\Pi(iy) = \sum_{m=0}^{M} \kappa_m \operatorname{Re} W\Big(\frac{hm+iy}{\eta}\Big) + \sum_{m=0}^{M} \kappa_m \operatorname{Re} W\Big(\frac{h(m+1)+iy}{\eta}\Big) = \sum_{m=0}^{M+1} \tilde\kappa_m \operatorname{Re} W\Big(\frac{hm+iy}{\eta}\Big).$$

The term $m = 0$ equals $\operatorname{Re} W(iy/\eta) \ge 0$ by Lemma 4.2(4), since $\tilde\kappa_0 = 1$. Let $m \ge 1$, $\xi_m = (hm+iy)/\eta$ and $Y = (y/h)^2$. Then

$$\operatorname{Re}\frac{w_0}{\xi_m} = \frac{\eta w_0}{h}\cdot\frac{m}{m^2+Y}, \qquad |\xi_m|^3 = \frac{h^3(m^2+Y)^{3/2}}{\eta^3}.$$

Since $\operatorname{Re}\xi_m \ge h^-/\eta^+$, Lemma 4.3 gives

$$|W_0(\xi_m)| \le \frac{C(h^-/\eta^+)\,\eta^3}{h^3(m^2+Y)^{3/2}} \le \frac{\eta w_0}{h}\cdot\frac{1}{m(m^2+Y)}\cdot\frac{C(h^-/\eta^+)(\eta^+)^2}{(h^-)^2 w_0} < \frac{\eta w_0}{h}\cdot\frac{\varepsilon/m}{m^2+Y}.$$

Here we used $(m^2+Y)^{1/2} \ge m$, $\eta \le \eta^+$, $h \ge h^-$, and the inequality $C(h^-/\eta^+)(\eta^+)^2 < \varepsilon (h^-)^2 w_0$ of check (C6). Hence $\tilde\kappa_m \operatorname{Re} W(\xi_m) \ge (\eta w_0/h)\,(m\tilde\kappa_m - \varepsilon|\tilde\kappa_m|/m)/(m^2+Y)$, whatever the sign of $\tilde\kappa_m$. Summing over m,

$$\Pi(iy) \ge \frac{\eta w_0}{h}\, B\Big(\frac{y^2}{h^2}\Big), \qquad B(Y) = \sum_{m=1}^{M+1} \frac{m\tilde\kappa_m - \varepsilon|\tilde\kappa_m|/m}{m^2+Y}. \tag{5.1}$$

Let $\widetilde B(Y) = B(Y)\prod_{m=1}^{M+1}(m^2+Y)$, a polynomial with rational coefficients. Check (C6) computes $\widetilde B$ and its Sturm sequence exactly. It shows $\widetilde B(0) > 0$ and equal numbers of sign variations at 0 and $+\infty$. By Sturm's theorem $\widetilde B$ has no zero in $[0,\infty)$, so $\widetilde B > 0$ and $B > 0$ there. Thus $\Pi(iy) \ge 0$ for all real y.

The line $\operatorname{Re} z = h$. Here $\Pi(h+iy) = \operatorname{Re} F(h+iy) + \operatorname{Re} F(iy) = \Pi(iy) \ge 0$.

Horizontal segments. Let $z = x + iy$ with $0 \le x \le h$ and $|y|$ large. By Lemma 4.3, $W(\xi) = w_0/\xi + O(|\xi|^{-3})$ uniformly for $\operatorname{Re}\xi \ge 0$. Since $\operatorname{Re}(\eta w_0/(a+iy)) = \eta w_0 a/y^2 + O(y^{-4})$ uniformly for $0 \le a \le (M+1)h$,

$$\Pi(x+iy) = \frac{\eta w_0}{y^2}\sum_{m=0}^{M}\kappa_m\big((x+hm)+(h-x+hm)\big) + O(|y|^{-3}) = \frac{\eta w_0 h}{y^2}\sum_{m=0}^{M}(2m+1)\kappa_m + O(|y|^{-3}),$$

uniformly in x. Check (C6) gives $\sum_m (2m+1)\kappa_m > 0$, so $\Pi(x+iy) > 0$ whenever $|y| \ge y_0$, for some y_0.

Conclusion. Π is harmonic on the open strip and continuous on its closure. On the boundary of each rectangle $[0,h]\times[-y_1,y_1]$ with $y_1\ge y_0$ it is nonnegative. By the minimum principle, $\Pi\ge 0$ inside each such rectangle, hence on the whole closed strip. □

Remark 5.4. In the proof of Lemma 5 of BTY [1], in our notation, the boundary value is bounded as $\Pi(iy)\ge \eta w_0 h\sum_{m\ge1}\tilde\kappa_m B_m(y)$. Here $B_m(y)=(m-\varepsilon_0/m)/(m^2+y^2)$ if $\tilde\kappa_m>0$ and $B_m(y)=(m+\varepsilon_0/m)/(c^2m^2+y^2)$ if $\tilde\kappa_m\le 0$, where $\varepsilon_0=1/2000$ is their tolerance and $c=151/153\le h$. The exact denominator is $h^2m^2+y^2=h^2(m^2+Y)$. Keeping the factor h^2 makes all denominators equal, so the loss from replacing h by 1 or by c disappears.

Lemma 5.5 (Paired transform off the strip)**.** *Let $h\in[h^-,1)$, $\eta\in(0,\eta^+]$, and let z satisfy $-\eta\le\operatorname{Re}z\le h+\eta$ and $|\operatorname{Im}z|\ge T_0$. Then $\Pi(z)\ge -S\eta/(\operatorname{Im}z)^2$, where*

$$S=2\|\kappa\|_1\Big(w_0+\frac{2C_-(\eta^+)^2}{T_0}\Big).$$

Proof. Write $z=x+iy$ and let x' be the point of $[0,h]$ nearest to x. Then $v=x-x'$ satisfies $|v|\le\eta<1$, and $\Pi(x'+iy)\ge 0$ by Proposition 5.3. Each of the $2(M+1)$ transforms in $\Pi(z)$ has the form $W((\xi+v')/\eta)$, and the corresponding transform in $\Pi(x'+iy)$ is $W(\xi/\eta)$. In $\operatorname{Re}F(z)$ the m-th term has $\xi=x'+hm+iy$ and $v'=v$. In $\operatorname{Re}F(h-\bar z)$ it has $\xi=h-x'+hm+iy$ and $v'=-v$. In both cases $\operatorname{Re}\xi\ge0$ and $\operatorname{Im}\xi=y$. Also $\operatorname{Re}((\xi+v')/\eta)\ge-1$, because $x\ge-\eta$ and $h-x\ge-\eta$. By Lemma 4.3,

$$\Big|W\Big(\frac{\xi+v'}{\eta}\Big)-W\Big(\frac{\xi}{\eta}\Big)\Big|\le\eta w_0\frac{|v'|}{|\xi|\,|\xi+v'|}+\frac{2C_-\eta^3}{|y|^3}\le\frac{\eta}{y^2}\Big(w_0+\frac{2C_-(\eta^+)^2}{T_0}\Big).$$

The last step uses $|\xi|,|\xi+v'|\ge|y|$, $|v'|<1$, $\eta\le\eta^+$ and $|y|\ge T_0$. Multiplying by $|\kappa_m|$ and summing over both halves of Π gives the claim. □

6. The explicit formula and its remainder

This section applies Kadiri's formula to the test function and shows that its remainder is a sum over the trivial zeros. For $\operatorname{Re}s>1/2$ we define Kadiri's remainder for the kernel w at scale η:

$$\Delta(s)=\frac{1}{2\pi i}\int_{(1/2)}W_0\Big(\frac{s-z}{\eta}\Big)\operatorname{Re}\psi\Big(\frac{z}{2}\Big)\,\mathrm{d}z+W_0\Big(\frac{s}{\eta}\Big). \tag{6.1}$$

The integral converges absolutely, by Lemma 4.3 with $\nu=(\operatorname{Re}s-1/2)/\eta$ and by the logarithmic growth of ψ (Lemma 6.2 below).

Lemma 6.1 (Explicit formula for f)**.** *For every s with $\operatorname{Re}s>1/2$,*

$$\operatorname{Re}\sum_{n\ge1}\frac{\Lambda(n)}{n^s}f(\log n)=\eta w_0\sum_{m=0}^{M}\kappa_m\Big(-\frac12\log\pi+\frac12\operatorname{Re}\psi\Big(\frac{s+hm}{2}+1\Big)\Big)$$
$$+\operatorname{Re}F(s-1)-\sum_{\rho}\operatorname{Re}F(s-\rho)+\sum_{m=0}^{M}\kappa_m\operatorname{Re}\Delta(s+hm).$$

Proof. Apply (K2) to $g=w_\eta$. Since (K2) asks for support in the half-open interval $[0,d')$ and w_η vanishes from d/η on, any $d'>d/\eta$ works. We take $d'=d/\eta+1$. By Lemma 4.2(3), w_η satisfies the hypotheses of (K2), and $w_\eta\ge0$. Its transform is $\mathcal{L}g(z)=W(z/\eta)$, and $g(0)=\eta w_0$. Integration by parts with $g'(0)=0$ gives $\mathcal{L}g''(z)=z^2\mathcal{L}g(z)-z\eta w_0$, so

$$\frac{\mathcal{L}g''(z)}{z^2}=W\Big(\frac{z}{\eta}\Big)-\frac{\eta w_0}{z}=W_0\Big(\frac{z}{\eta}\Big).$$

Hence the last line of (2.1) is $\operatorname{Re}\Delta(s)$. Apply (2.1) at $s+hm$, multiply by κ_m and sum over m. On the left, $\sum_m \kappa_m n^{-hm} w_\eta(\log n) = f(\log n)$. On the right, $\sum_m \kappa_m W((s+hm-v)/\eta) = F(s-v)$ for $v=1$ and $v=\rho$. □

Lemma 6.2 (Growth of ψ)**.** *For each $\alpha>0$, $|\psi(z)| = O_\alpha(\log(2+|z|))$ uniformly for $\operatorname{Re} z \ge \alpha$.*

Proof. Use the series in (D). For $k \le 2|z|+2$, bound the two summands separately using $|k+z| \ge k+\alpha$. Their sum is $O_\alpha(\log(2+|z|))$. For $k > 2|z|+2$, combine them as $(z-1)/((k+1)(k+z))$. This has modulus at most $2(|z|+1)/(k+1)^2$, so the sum over these k is $O(1)$. □

Proposition 6.3 (Remainder over the trivial zeros)**.** *Let $\eta>0$ and $\operatorname{Re} s > 1/2$. Then*

$$\Delta(s) = -\sum_{n=1}^{\infty} W_0\Big(\frac{s+2n}{\eta}\Big), \tag{6.2}$$

and the series converges absolutely.

Proof. A meromorphic weight. On the line $\operatorname{Re} z = 1/2$ we have $\bar z = 1-z$ and $\psi(\bar w) = \overline{\psi(w)}$. Hence $\operatorname{Re}\psi(z/2) = \Psi(z)$ there, where

$$\Psi(z) = \frac{1}{2}\Big(\psi\Big(\frac{z}{2}\Big) + \psi\Big(\frac{1-z}{2}\Big)\Big)$$

is meromorphic on $\mathbb{C}$. By (D), $\psi(z/2)$ has poles at $z = 0, -2, -4, \dots$ with residue -2, so Ψ has residue -1 there. The poles of $\psi((1-z)/2)$ are at $z = 1, 3, 5, \dots$, to the right of the line.

The other factor. Put $V(z) = W_0((s-z)/\eta)$ and $\nu = (\operatorname{Re} s - 1/2)/\eta > 0$. The only possible pole of V is at $z=s$, to the right of the line. For $\operatorname{Re} z \le 1/2$ we have $\operatorname{Re}((s-z)/\eta) \ge \nu$, so Lemma 4.3 gives $|V(z)| \le C(\nu)\eta^3 |s-z|^{-3}$.

Contour. Fix an integer $N \ge 0$ and $Y > |\operatorname{Im} s| + 1$. Integrate $V\Psi$ counterclockwise around the rectangle with vertices $\frac12 \pm iY$ and $-2N-1 \pm iY$. Its right side is the line segment in (6.1), traversed upwards. Inside lie the poles $z = -2n$, $0 \le n \le N$, and

$$\frac{1}{2\pi i}\oint V\Psi \,\mathrm{d}z = -\sum_{n=0}^{N} V(-2n) = -\sum_{n=0}^{N} W_0\Big(\frac{s+2n}{\eta}\Big).$$

Horizontal sides. Let $z = x \pm iY$ with $-2N-1 \le x \le 1/2$. By (D), $\psi(z/2) = \psi(z/2+N+1) - \sum_{l=0}^{N}(z/2+l)^{-1}$, and $\operatorname{Re}(z/2+N+1) \ge 1/2$. Each term of the sum has modulus at most $2/Y$. Also $\operatorname{Re}((1-z)/2) \ge 1/4$. By Lemma 6.2, $|\Psi(z)| = O_N(\log Y)$ on these sides. Since $|V(z)| = O(Y^{-3})$ there, the horizontal integrals tend to 0 as $Y \to \infty$.

Left side. Let $z = -2N-1+iy$. By (D), $\psi(z/2) = \psi(\frac12 + \frac{iy}{2}) - \sum_{l=0}^{N}(-N-\frac12+l+\frac{iy}{2})^{-1}$. The sum has modulus at most $\sum_{r=0}^{N}(r+\frac12)^{-1} = O(\log(N+2))$. The other argument $(1-z)/2 = N+1-iy/2$ has real part at least 1. By Lemma 6.2, $|\Psi(z)| = O(\log(N+2) + \log(2+|y|))$, uniformly in N. Put $X = \operatorname{Re} s + 2N + 1$. Then $|s-z|^2 = X^2 + (\operatorname{Im} s - y)^2$. The substitution $y = \operatorname{Im} s + Xv$ shows that the integral over the left side is $O\big((\log(N+2) + \log X)/X^2\big)$. This tends to 0 as $N \to \infty$.

Conclusion. Let $Y \to \infty$ and then $N \to \infty$. The integral in (6.1) equals $-\sum_{n\ge0} W_0((s+2n)/\eta)$. The series converges absolutely, since $|W_0((s+2n)/\eta)| \le C(\nu)\eta^3|s+2n|^{-3}$. The term $n=0$ cancels the term $W_0(s/\eta)$ in (6.1), which proves (6.2). □

Remark 6.4. Heuristically, the trivial zeros $-2n$ contribute $-\sum_n W((s+2n)/\eta)$ to the explicit formula, just as the nontrivial zeros contribute $-\sum_\rho W((s-\rho)/\eta)$. This series diverges. By (D), the digamma term in (2.1) is a regularized form of its main part $-\sum_n \eta w_0/(s+2n)$. Identity (6.2) says that $\Delta(s)$ is exactly the convergent rest. For their kernel, Lemma 8 of BTY [1] bounds $|\Delta(s)|$ by $623\eta^3$ when $\operatorname{Im} s = 0$ and by $14\eta^2 + 424\eta^3$ when $\operatorname{Im} s \ge H$. They

integrate the absolute value of the integrand in (6.1). The η^2 term comes from a factor $\log \operatorname{Im} s$ together with their assumption $\eta \log t \le A_0$, the analogue of $\eta L \le A_*$. By (6.2), $\Delta(s)$ is $O(\eta^3)$ uniformly in the height.

Lemma 6.5 (Remainder bound)**.** *For $0 \le m \le M$ put $u_m = \sigma_m^- + 2$ and $C_m = C(u_m/\eta^+)$, and define*

$$E = \sum_{m=0}^{M} |\kappa_m|\, C_m \Big\{ \frac{1}{u_m^3} + \frac{1}{4u_m^2} + a_+\Big(\frac{1}{H^3} + \frac{1}{2H^2}\Big)\Big\},$$

with a_+ from (3.2). Then $\sum_{k=0}^{K} a_k \sum_{m=0}^{M} |\kappa_m|\, |\Delta(s_k + hm)| \le E\eta^3$.

Proof. Fix k and m and put $s = s_k + hm$. Then $\operatorname{Re} s = \sigma + hm \ge \sigma_m^-$ by Lemma 3.2. For $n \ge 1$, $\operatorname{Re}((s+2n)/\eta) \ge u_m/\eta^+$, so Lemma 4.3 and Proposition 6.3 give

$$|\Delta(s)| \le C_m \eta^3 \sum_{n \ge 1} |s + 2n|^{-3}.$$

For a positive decreasing function ϕ on $[x_1, \infty)$, $\sum_{n\ge1} \phi(x_1 + 2(n-1)) \le \phi(x_1) + \frac{1}{2}\int_{x_1}^{\infty} \phi$. If $k = 0$, then $|s + 2n| \ge \sigma_m^- + 2n$. Taking $\phi(x) = x^{-3}$ and $x_1 = u_m$ gives at most $u_m^{-3} + 1/(4u_m^2)$. If $k \ge 1$, then $|s+2n|^2 \ge (\sigma_m^- + 2n)^2 + \tau^2$ with $\tau = kt > H$. Take $\phi(x) = (x^2 + \tau^2)^{-3/2}$ and use $\int_0^{\infty} \phi = \tau^{-2}$ to get at most $H^{-3} + 1/(2H^2)$. Sum with the weights $a_0 = 1$ and $\sum_{k\ge1} a_k = a_+$. □

7. The terms of the explicit formula

This section bounds, in turn, the zeros, the gamma factor, the pole and the primes. Throughout, the hypotheses of Section 3 hold and j is fixed.

7.1. **Zeros.** For $\tau > 0$ put $\phi_\tau(x) = (x - \tau)^{-2} + (x + \tau)^{-2}$. The constant Z_0 below bounds the loss from zeros above the cutoff T in the sum for $k = 0$, and Z_+ does so for each $k \ge 1$. The numbers $\lambda_1, \lambda_2, \lambda_3$ collect the logarithms that appear when (L1) is applied. Define the band constants

$$\lambda_1 = \log \frac{K + T_0/H}{2\pi}, \qquad \lambda_2 = \log\Big(K + \frac{T_0}{H}\Big), \qquad \lambda_3 = \frac{\log \frac{H+T_0}{2\pi}}{2H + T_0} + \frac{\log \frac{2H+T_0}{T_0}}{H},$$

$$Z_0 = \frac{2S\eta^+ \log T_0}{T_0},$$

$$Z_+ = SA_*\Big\{ \frac{1 + \lambda_1/\log H}{2\pi T_0} + \frac{\lambda_3}{2\pi \log H} + \frac{8(1 + \lambda_2/\log H)}{T_0^2} + \frac{4}{HT_0 \log H}\Big\},$$

and $Z = Z_0 + a_+ Z_+$. Check (C7) gives $Z_0, Z_+ < 10^{-8}$.

Lemma 7.1 (Zero sums)**.** *For $0 \le k \le K$,*

$$\sum_{\rho} \operatorname{Re} F(s_k - \rho) \ge \begin{cases} -Z_0, & k = 0, \\ F(\sigma - 1 + \eta) + F(\sigma - \eta) - Z_+, & k = 1, \\ -Z_+, & 2 \le k \le K. \end{cases}$$

Proof. Pairing. The multiset of zeros is invariant under $\rho \mapsto 1 - \bar\rho$, and so are its parts with $|\gamma| \le T$ and $|\gamma| > T$. A direct computation gives $s_k - (1 - \bar\rho) = h - \overline{(s_k - \rho)}$. Hence, over either part,

$$\sum_{\rho} \operatorname{Re} F(s_k - \rho) = \frac{1}{2} \sum_{\rho} \Pi(s_k - \rho).$$

Both sums converge absolutely, since $\operatorname{Re} F(z) = O(|\operatorname{Im} z|^{-2})$ by Lemma 4.3.

Zeros with $|\gamma| \le T$. By Lemma 3.3, $\operatorname{Re}(s_k - \rho) = \sigma - \beta \in [0, h]$, so each $\Pi(s_k - \rho) \ge 0$ by Proposition 5.3. For $k = 1$, the zeros ρ_0 and $1 - \bar\rho_0$ are distinct, because $\beta_0 = 1 - \eta \ne 1/2$. For both of them $\Pi(s_1 - \rho) = F(\sigma - \beta_0) + F(\sigma - 1 + \beta_0) = F(\sigma - 1 + \eta) + F(\sigma - \eta)$, with real arguments. Keeping these two terms and dropping the others gives the main term for $k = 1$.

Zeros with $|\gamma| > T$. Since $0 < \beta < 1$, we have $\operatorname{Re}(s_k - \rho) = \sigma - \beta \in (\sigma - 1, \sigma)$. Since $\sigma - 1 = -\mu\eta$ and $\sigma = h + \mu\eta$, this interval lies in $(-\eta, h + \eta)$. Also $|\operatorname{Im}(s_k - \rho)| = |kt - \gamma| > T - Kt = T_0$. Lemma 5.5 gives

$$\sum_{|\gamma|>T} \operatorname{Re} F(s_k - \rho) \ge -\frac{S\eta}{2} \sum_{|\gamma|>T} \frac{1}{(kt - \gamma)^2}.$$

For $k = 0$ the last sum is $2\sum_{\gamma>T} \gamma^{-2} < 2T_0^{-1} \log T_0$, by (L2) and since $x^{-1}\log x$ decreases for $x \ge e$. Hence the loss is at most $S\eta^+ T_0^{-1} \log T_0 = Z_0/2$.

For $k \ge 1$ put $\tau = kt$, so that the last sum is $\sum_{\gamma>T} \phi_\tau(\gamma)$. The function ϕ_τ is continuous, positive and decreasing on $[T, \infty)$, and $T > 2\pi e$. Apply (L1) with $T_1 = T$ and let $T_2 \to \infty$. Then

$$\begin{aligned}\int_T^\infty \phi_\tau(x) \log \frac{x}{2\pi}\,\mathrm{d}x &\le \int_{\tau+T_0}^\infty \phi_\tau(x) \log \frac{x}{2\pi}\,\mathrm{d}x \\ &= \Big(\frac{1}{T_0} + \frac{1}{2\tau + T_0}\Big) \log \frac{\tau + T_0}{2\pi} + \frac{1}{\tau} \log \frac{2\tau + T_0}{T_0} \le \frac{L + \lambda_1}{T_0} + \lambda_3.\end{aligned}$$

The first step enlarges the range of integration, since $T \ge \tau + T_0$ and the integrand is positive. The equality is an integration by parts with the antiderivative $-(x - \tau)^{-1} - (x + \tau)^{-1}$ of ϕ_τ. For the last step, $(\tau + T_0)/(2\pi) \le e^L(K + T_0/H)/(2\pi)$ bounds the first term. The second and third terms are at most their values at $\tau = H$, whose sum is λ_3, because they decrease in $\tau \ge H$. For the second term, the derivative of $\log((\tau + T_0)/2\pi)/(2\tau + T_0)$ has the sign of $(2\tau + T_0)/(\tau + T_0) - 2\log((\tau + T_0)/2\pi)$, which is negative because $(\tau + T_0)/2\pi > e$. For the third term, $\tau^{-1}\log(1 + 2\tau/T_0)$ decreases because $x^{-1}\log(1 + x)$ decreases for $x > 0$. For the error terms of (L1), $\phi_\tau(T) \le 2/T_0^2$, $\log T \le L + \lambda_2$, and

$$\int_T^\infty \frac{\phi_\tau(x)}{x}\,\mathrm{d}x \le \frac{1}{T}\Big(\frac{1}{T - \tau} + \frac{1}{T + \tau}\Big) \le \frac{2}{HT_0}.$$

Insert $\eta = A/L$, $A \le A_*$ and $L > \log H$, and note that λ_1, λ_2, λ_3 are positive. The loss is then at most $Z_+/2$. □

Remark 7.2. The constants Z_0 and Z_+ are twice what the proof needs, because the certificate does not use the factor $1/2$ from the pairing. We keep the certificate's constants so that every constant in this paper agrees with the program.

7.2. **Gamma factor.** For $0 \le m \le M$ put $x_m = (\sigma^+ + h^+ m)/2 + 1$ if $\kappa_m > 0$ and $x_m = \sigma_m^-/2 + 1$ otherwise. Define

$$\begin{aligned}G = -\Big\{\frac{1}{2}\sum_{m=0}^{M} \kappa_m \psi(x_m) - \frac{\kappa(1)}{2}\log\pi + \frac{\kappa(1)}{2}\sum_{k=1}^{K} a_k \log k \\ - \frac{a_+\kappa(1)}{2}\log(2\pi) + \frac{a_+\|\kappa\|_1}{2}\cdot 10^{-10}\Big\},\end{aligned}$$

and $Q = a_+\kappa(1)w_0/2$. Check (C8) gives $G > 0$.

Lemma 7.3 (Gamma factor)**.**

$$\eta w_0 \sum_{k=0}^{K} a_k \sum_{m=0}^{M} \kappa_m\Big(-\frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{s_k+hm}{2}+1\Big)\Big) \le QA - Gw_0\eta.$$

Proof. The term $k=0$. The argument $(\sigma+hm)/2+1$ is real and lies between $\sigma_m^-/2+1$ and $(\sigma^+ + h^+m)/2+1$ by Lemma 3.2. Since ψ increases on $(0,\infty)$, $\kappa_m\psi((\sigma+hm)/2+1) \le \kappa_m\psi(x_m)$ for every m.

The terms $k \ge 1$. Here the argument is $x+iy$ with $x = (\sigma+hm)/2+1 \in (1,(M+3)/2]$ and $y = kt/2 \ge H/2 > x$. By (K3),

$$\Big|\operatorname{Re}\psi(x+iy) - \log\frac{kt}{2}\Big| \le \frac{(M+3)/2}{2(H/2)^2} + \frac{1}{12(H/2)} + \frac{((M+3)/2)^2}{2(H/2)^2} < 10^{-10},$$

by check (C8). Hence $\sum_m \kappa_m \frac{1}{2}\operatorname{Re}\psi(\cdot) \le \frac{\kappa(1)}{2}(L + \log k - \log 2) + \frac{\|\kappa\|_1}{2}10^{-10}$.

Sum. Multiply by a_k, add the terms $-\frac{1}{2}\kappa(1)a_k\log\pi$, and sum over k with $a_0 = 1$. The coefficient of L is $a_+\kappa(1)/2$, and the constant term is $-G$. Multiplying by ηw_0 and using $\eta L = A$ gives the claim. □

7.3. **The pole.**

Lemma 7.4 (Nonreal pole terms)**.** $\sum_{k=1}^{K} a_k \operatorname{Re} F(s_k - 1) \le 10^{-10}\eta$.

Proof. Let $k \ge 1$ and $\xi = (\sigma - 1 + hm + ikt)/\eta$. Then $x = \sigma - 1 + hm$ satisfies $|x| \le M$, and $\operatorname{Re}\xi \ge (\sigma-1)/\eta = -\mu > -1$. Hence $|\operatorname{Re}(w_0/\xi)| = w_0\eta|x|/(x^2+k^2t^2) \le Mw_0\eta/H^2$. By Lemma 4.3, $|W_0(\xi)| \le C_-\eta^3/H^3 \le C_-(\eta^+)^2\eta/H^3$. Summing over m and k gives at most $a_+\|\kappa\|_1(Mw_0/H^2 + C_-(\eta^+)^2/H^3)\,\eta$, which is below $10^{-10}\eta$ by check (C9). □

7.4. **Primes.** Let $\mathcal{P}$ be the set of primes up to 19 in bands 1–6 and up to 97 in band 7. The certified margins in Table 2 decrease with the height and are smallest in band 7, which ends where Yang's region takes over (Lemma 3.1). This is why band 7 uses a multiplier of degree 14 and more prime powers. Let $r_p = 10$ in bands 1–6 and $r_p = 20$ in band 7. For $p \in \mathcal{P}$ and $1 \le r \le r_p$ define the prime weight and the prime phase function

$$\omega_{p,r} = \kappa(p^{-rh^-})\,\frac{w(\eta^+ r\log p)}{w_0}\,p^{-r\sigma^+}, \qquad \Omega_p(x) = \sum_{r=1}^{r_p}\omega_{p,r}P(rx).$$

Let J_p be a certified lower bound for $\min_x \Omega_p(x)$ from check (C12), and put $J = \sum_{p\in\mathcal{P}} J_p\log p$.

Lemma 7.5 (Prime powers)**.** $\sum_{n\ge1}\Lambda(n)n^{-\sigma}f(\log n)P(t\log n) \ge \eta w_0 J$.

Proof. Every term of the sum is nonnegative, by Lemma 5.2, $P \ge 0$ and $\Lambda \ge 0$. We keep only the terms $n = p^r$ with $p \in \mathcal{P}$ and $r \le r_p$. These n are distinct. For such n,

$$n^{-\sigma}f(\log n) = \eta\, w(\eta\log n)\,\kappa(n^{-h})\,n^{-\sigma} \ge \eta\, w(\eta^+\log n)\,\kappa(n^{-h^-})\,n^{-\sigma^+} = \eta w_0\,\omega_{p,r}.$$

The three replacements use that w is nonincreasing and $\eta \le \eta^+$, that κ is decreasing and $n^{-h} \le n^{-h^-}$, and that $\sigma \le \sigma^+$. All factors are nonnegative. Since $\Lambda(p^r) = \log p$, the kept terms are at least $\eta w_0 \sum_p \log p\,\Omega_p(t\log p) \ge \eta w_0 J$. □

For comparison, Lemmas 10 and 11 of BTY [1] show $f(u) \ge \kappa(1)f(0)$ for $0 \le u \le 59$ and bound the same sum below by $\kappa(1)f(0)\sum_{p<100} m_p\log p$. Here m_p is a certified lower bound for $\min_x \sum_{r=1}^{15} p^{-r}P(rx)$. BTY show that this sum, with $p^{-r\sigma}$ in place of p^{-r}, decreases in σ. Our weights $\omega_{p,r}$ keep the decay of w, of κ and of $n^{-\sigma}$ for each prime power separately.

8. The final inequality

This section combines the bounds into one inequality for (L, A) and proves Theorem 1.1. Define the zero-pair term

$$D = a_1\{F(\sigma - 1 + \eta) + F(\sigma - \eta)\} - F(\sigma - 1).$$

Proposition 8.1 (Contradiction inequality)**.** *The minimal zero satisfies*

$$D + (G + J)w_0\eta - E\eta^3 - Z - 10^{-10}\eta - QA \le 0. \tag{8.1}$$

Proof. Since $P(t\log n) = \operatorname{Re}\sum_k a_k n^{-ikt}$ and f is real,

$$\sum_{n\ge1}\Lambda(n)n^{-\sigma}f(\log n)P(t\log n) = \sum_{k=0}^{K} a_k \operatorname{Re}\sum_{n\ge1}\frac{\Lambda(n)}{n^{s_k}}f(\log n).$$

Apply Lemma 6.1 at each s_k. All of them have real part $\sigma > 1/2$. Lemma 7.5 bounds the left side below by $\eta w_0 J$. On the right, Lemma 7.3 bounds the gamma terms by $QA - Gw_0\eta$. The pole terms are $F(\sigma - 1)$ for $k = 0$ plus at most $10^{-10}\eta$ by Lemma 7.4. By Lemma 7.1, the zero terms are at most $-a_1\{F(\sigma - 1 + \eta) + F(\sigma - \eta)\} + Z$. By Lemma 6.5, the remainders are at most $E\eta^3$. Together,

$$\eta w_0 J \le QA - Gw_0\eta + F(\sigma - 1) - a_1\{F(\sigma - 1 + \eta) + F(\sigma - \eta)\} + Z + 10^{-10}\eta + E\eta^3,$$

which is (8.1). □

8.1. **Evaluation of the zero-pair term.** Substituting $\sigma = 1 - \mu\eta$ and $h = 1 - 2\mu\eta$ into the definition of F gives

$$D = D_0(\mu) + a_1\sum_{m=0}^{M}\kappa_m W(z_m^{(1)}) + \sum_{m=1}^{M}\kappa_m(a_1 W(z_m^{(2)}) - W(z_m^{(3)})), \tag{8.2}$$

where $D_0(\mu) = a_1 W(1 - \mu) - W(-\mu)$ and

$$z_m^{(1)} = \frac{m+1}{\eta} - (2m+1)\mu - 1, \qquad z_m^{(2)} = \frac{m}{\eta} - (2m+1)\mu + 1, \qquad z_m^{(3)} = \frac{m}{\eta} - (2m+1)\mu.$$

These are the arguments of $F(\sigma - \eta)$, $F(\sigma - 1 + \eta)$ and $F(\sigma - 1)$. For $m = 0$ the last two are $1 - \mu$ and $-\mu$. Since $\eta < \eta^+$ and $\mu < 1$, the smallest argument in the two sums of (8.2) is $z_1^{(3)} = 1/\eta - 3\mu$, so all of them exceed $1/\eta^+ - 3 > 0$. The program evaluates them by Lemma 4.4. For the small arguments, define the moments

$$I_n = \int_0^d u^n(a_1 - e^u)w(u)\,\mathrm{d}u \qquad (0 \le n \le 8).$$

Lemma 8.2 (Small arguments)**.** *If $9/10 < \mu < 1$, then $D_0(\mu) \ge \sum_{n=0}^{8}(\mu - 1)^n I_n/n! - 10^{-12}$.*

Proof. We have $D_0(\mu) = \int_0^d e^{-(1-\mu)u}(a_1 - e^u)w(u)\,\mathrm{d}u$. By Taylor's theorem,

$$\Big|e^{-x} - \sum_{n=0}^{8}\frac{(-x)^n}{n!}\Big| \le \frac{x^9}{9!} \qquad (x \ge 0).$$

With $x = (1 - \mu)u \le u/10$, $0 \le w \le w_0$ and $|a_1 - e^u| \le a_1 + e^d$, the error is at most

$$\frac{w_0(a_1 + e^d)d^{10}}{10 \cdot 9! \cdot 10^9} < 10^{-12},$$

by check (C11). □

8.2. **Proof of Theorem 1.1.** For the band j and $(L, A) \in U_j$, let $\mu = L/\log(Ke^L + T_0)$ and $\eta = A/L$, and define the margin

(8.3)
$$\Theta_j(L,A) = \sum_{n=0}^{8} \frac{(\mu-1)^n I_n}{n!} - 10^{-12} + a_1 \sum_{m=0}^{M} \kappa_m W(z_m^{(1)}) + \sum_{m=1}^{M} \kappa_m (a_1 W(z_m^{(2)}) - W(z_m^{(3)})) + (G+J)w_0\eta - E\eta^3 - Z - 10^{-10}\eta - QA.$$

All constants here belong to band j, and W is given by (4.2).

Proposition 8.3 (Certificate)**.** *For $1 \le j \le 7$ and every $(L, A) \in U_j$, $\Theta_j(L, A) > 10^{-6}$.*

Proof. This is check (C13) of Section 9. □

Proof of Theorem 1.1. Suppose the theorem is false. Lemma 3.1 gives a minimal zero ρ_0 with $(L, A) \in U_j$ for some j. Lemma 3.2 gives $9/10 < \mu < 1$. By Proposition 8.1, (8.2) and Lemma 8.2, $\Theta_j(L, A) \le 0$. This contradicts Proposition 8.3. □

The threshold 10^{-6} is a safety margin of the program. The proof needs only $\Theta_j > 0$. The minimal zero may lie on the boundary $A = A_*$, so the region in Theorem 1.1 is closed.

9. What the computer checks

9.1. **Arithmetic.** The program uses python-flint 0.8.0 [12], which wraps FLINT 3.3.1 and its Arb module [11, 3]. Every input decimal is read as an exact rational. Exact statements (kernel coefficients, a_k, Bernstein coefficients, Sturm sequences) use rational arithmetic. All other quantities are Arb balls at 160-bit precision. Every operation encloses the true value. An inequality counts as proved only when it holds for the whole ball. Integrals are computed with Arb's rigorous integration [4], with absolute tolerance 10^{-30}. Floating-point numbers are used only to measure the running time.

9.2. **The finite statements.** The program checks the following statements. The lemma that uses each one is given in brackets.

- (C1) $L_0 < \log H$, $\log(L_7)/21.233 > A_*$, adjacent bands share their endpoints exactly, and $1 - 1/(R\log 2) > 1/2$ [Lemma 3.1].
- (C2) The exact coefficients of w on $[0, d]$, the negative Bernstein coefficients of $\mathrm{d}q/\mathrm{d}(y^2)$, the identities and signs of Lemma 4.2(3), and the exact value of w_4^* [Lemma 4.2].
- (C3) $a_0 = 1$ and $a_k \ge 0$ for all k, exactly [Section 7].
- (C4) For each band, $\kappa_0 = 1$, $\kappa(1) > 0$, and the degree-M Bernstein coefficients of κ are positive and strictly decreasing [Lemma 5.2].
- (C5) $1 - A_*/\log(Ke^{L_{j-1}} + T_0) > \sigma^-$, $A_*/L_{j-1} < \eta^+$, $0.99 < \sigma^- < \sigma^+ < 1$ and $\eta^+ < 0.01$ [Lemma 3.2].
- (C6) The inequalities $0 < \varepsilon < 1$, $C(h^-/\eta^+)(\eta^+)^2 < \varepsilon(h^-)^2 w_0$ and $\sum_m (2m+1)\kappa_m > 0$, and the Sturm test for $\widetilde{B}(Y)$ [Proposition 5.3].
- (C7) $Z_0 < 10^{-8}$ and $Z_+ < 10^{-8}$ [Lemma 7.1].
- (C8) The inequality $(M+3)/2 < H/2$, the bound 10^{-10} for $|\operatorname{Re}\psi(x+iy) - \log(kt/2)|$ in the proof of Lemma 7.3, and $G > 0$ [Lemma 7.3].
- (C9) $a_+\|\kappa\|_1(Mw_0/H^2 + C_-(\eta^+)^2/H^3) < 10^{-10}$ [Lemma 7.4].
- (C10) $\sigma^- > 1/2$, and E is finite and positive [Lemma 6.5].
- (C11) Enclosures of $I_0, \dots, I_8$ with radius below 10^{-25}, and the Taylor bound of Lemma 8.2 [Lemma 8.2].

(C12) For each $p \in \mathcal{P}$, primality, distinctness, $0 < \eta^+ r_p \log p < d$, $w(\eta^+ r \log p) > 0$ and $\kappa(p^{-rh^-}) > 0$. Also the nonnegativity of the cosine coefficients of Ω_p and the cell bounds described below [Lemma 7.5].

(C13) $\Theta_j > 10^{-6}$ on every box of a partition of U_j. On every box also $9/10 < \mu < 1$, $0 < \eta < \eta^+$, $\sigma > \sigma^-$, and all arguments $z_m^{(i)}$ in (8.2) exceed $1/\eta^+ - 3$ [Proposition 8.3].

9.3. **Covering the band rectangles.** Each band rectangle U_j is first cut into 128×16 equal closed boxes with rational corners. For each box, the program encloses its two sides in Arb balls and evaluates Θ_j on them. By the inclusion property of ball arithmetic, the result encloses Θ_j on the whole box. A box passes if the lower endpoint exceeds 10^{-6}. Otherwise it is bisected into two closed halves, alternately in A and in L, and both halves are checked. A box of depth 40 that fails stops the program with an error. At the end the program checks that the number of passed boxes equals 2048 plus the number of bisections, so no box is lost. Table 2 reports the counts.

9.4. **Prime phase functions.** The function $\Omega_p(x) = \sum_n \hat{\Omega}_p(n) \cos(nx)$ has coefficients $\hat{\Omega}_p(n) = \sum_{kr=n} \omega_{p,r} a_k \geq 0$. It is even and 2π-periodic, so its minimum is attained on $[0, \pi]$. The program cuts $[0, \pi]$ into equal closed cells of radius δ, with 1024 cells in bands 1–6 and 2048 cells in band 7. At the centre θ of each cell it encloses $\Omega_p(\theta)$ and $\Omega_p'(\theta)$ by Horner evaluation at $e^{i\theta}$. Taylor's theorem gives, on the whole cell,

$$\Omega_p(x) \geq \Omega_p(\theta) - |\Omega_p'(\theta)|\, \delta - \frac{\delta^2}{2} \sum_n n^2 \hat{\Omega}_p(n).$$

The number J_p is the smallest lower endpoint of these bounds over all cells, and the program requires $J_p > 0$. In total 100352 cells are checked.

9.5. **Command and running time.** The code is at `https://github.com/ArseniyVarlamov/zeta-zero-free-region`, commit `7235b86064b6`. It consists of a driver `verify.py`, the band checker `verify_band.py`, the exact kernel construction `polynomial_kernel.py`, the input file `parameters.json`, and a wrapper `verify.sh`. The command `bash verify.sh` installs python-flint 0.8.0 in a local environment if needed and prints a JSON record with `"status": "PASS"`. The record contains every certified constant of Tables 1 and 2. On one core of an Apple M3 Pro the run takes about 19 seconds and uses about 33 MB of memory. Band 7 takes 13 of these seconds.

9.6. **What is not checked by the program.** The program does not re-prove the inputs (K1)–(K3), (PT), (Y), (L1), (L2) and (D). It also does not check the analytic arguments of Sections 3–8, which are proved in the text. FLINT and Arb are trusted. A Lean 4 formalization, with these inputs as hypotheses, is in the same repository.

Use of AI tools

AI tools were used in the computations, the Lean formalization and the writing.

Appendix A. Parameters

A.1. **Kernel.** The constants of Definition 4.1 are $b = -0.12045365$ and $d = 1.05396486$. Rounded to six significant digits, the polynomial piece of w on $[0, d]$ is

$$w(u) \approx 10 - 44.5499\, u^2 + 34.1907\, u^3 + 19.7176\, u^4 - 21.4917\, u^5 + 2.20979\, u^7 - 0.0712860\, u^9.$$

The program uses the exact rational coefficients. Rounded, $w''(0) = -89.0998$, $w'''(0) = 205.144$, $w_4^* = 797.744$ and $C_- = 3251.84$.

A.2. **Trigonometric polynomial.** The integers in (3.2) are

$$\begin{aligned}(c_0, \dots, c_{12}) = (&3107391881,\ -5311531843,\ -1551584903,\ 17515944216,\ 5809729718,\\ &-38311065182,\ -14579538641,\ 172588837119,\ 453058678886,\\ &619311376818,\ 537000910627,\ 291152319789,\ 78762658700),\end{aligned} \tag{A.1}$$

with $\sum_l c_l^2 = 10000000000005298367023 35$. They are the same in all bands.

A.3. **Bands.** Table 1 lists the band constants. Tables 3 and 4 list the multipliers. Table 2 lists the certified constants and the size of the computation.

TABLE 1. Band constants. All entries are exact. The last column gives the prime set $\mathcal{P}$, the exponent bound r_p and the number of cells per prime.

j	$[L_{j-1}, L_j]$	M	σ^-	η^+	ε	primes, r_p, cells
1	[28.7296, 45]	10	0.99334529	0.00723036	0.00048676	$p \le 19$, 10, 1024
2	[45, 60]	10	0.99562544	0.00461612	0.00019535	$p \le 19$, 10, 1024
3	[60, 70]	10	0.99667558	0.0034621	0.00010913	$p \le 19$, 10, 1024
4	[70, 77]	10	0.99713422	0.00296751	0.00007994	$p \le 19$, 10, 1024
5	[77, 80]	10	0.9973866	0.00269774	0.00006596	$p \le 19$, 10, 1024
6	[80, 82]	10	0.99748165	0.00259658	0.00006107	$p \le 19$, 10, 1024
7	[82, 82.32067]	14	0.99754126	0.00253325	0.0000581	$p \le 97$, 20, 2048

TABLE 2. Certified constants, rounded in the safe direction (G, J and Θ down, E, Z and S up), and the size of the computation. The last column is the smallest certified lower bound of Θ_j over the final boxes. Boxes are split only until they pass, so this number is close to 10^{-6} in bands where boxes were split.

j	G	J	E	Z	S	boxes	bisections	$\min \Theta_j \ge$
1	1.55128	0.228449	12.2969	$5.02 \cdot 10^{-9}$	111.081	2048	0	$4.15 \cdot 10^{-2}$
2	1.55345	0.228365	12.3153	$3.78 \cdot 10^{-9}$	114.033	2048	0	$1.72 \cdot 10^{-2}$
3	1.55436	0.228273	12.4803	$3.43 \cdot 10^{-9}$	123.445	2048	0	$6.89 \cdot 10^{-3}$
4	1.55474	0.228187	12.6207	$3.30 \cdot 10^{-9}$	129.117	2048	0	$1.20 \cdot 10^{-3}$
5	1.55507	0.228165	12.6286	$3.15 \cdot 10^{-9}$	129.794	2102	54	$9.43 \cdot 10^{-6}$
6	1.55516	0.228141	12.6299	$3.10 \cdot 10^{-9}$	129.980	2811	763	$1.75 \cdot 10^{-6}$
7	1.55533	0.228964	12.4744	$2.90 \cdot 10^{-9}$	123.289	16627	14579	$1.00012 \cdot 10^{-6}$

TABLE 3. Multiplier coefficients κ_m for bands 1–4 (exact).

m	band 1	band 2	band 3	band 4
0	1	1	1	1
1	−0.999856412099	−0.999908741467	−0.999959140973	−0.999977779994
2	0.996202231962	0.997343300509	0.998687184651	0.999234426187
3	−0.964429304272	−0.972999865470	−0.985023741291	−0.990703964052
4	0.832378658296	0.862899627312	0.913240158076	0.942777849286
5	−0.538465791838	−0.594616405027	−0.702460268613	−0.791028672628
6	0.188120516885	0.235924338084	0.342182468893	0.505196913937
7	0	0	0	−0.187383296683
8	0	−0.033275386754	−0.131280428384	0
9	−0.024245883442	0	0.082013165569	0.030702914887
10	0.010310013107	0.004640059874	−0.017396345470	−0.008816592212

TABLE 4. Multiplier coefficients κ_m for bands 5–7 (exact). Band 7 has degree 14.

m	band 5	band 6	band 7
0	1	1	1
1	−0.999981121591	−0.999981868859	−0.999978588421
2	0.999325039686	0.999347663974	0.999193002543
3	−0.991531939227	−0.991748468269	−0.989319433553
4	0.946405830390	0.947379889067	0.929535157749
5	−0.799626250262	−0.801972486575	−0.731139303640
6	0.516325105649	0.519394079858	0.359802841935
7	−0.194138362405	−0.196014531499	0
8	0	0	−0.097801312287
9	0.032774736667	0.033355146165	0
10	−0.009551506543	−0.009757975264	0.037296615268
11			0
12			0
13			−0.013964316125
14			0.006376973401

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